1. Plotting and Fitting of Binomial Distribution and Graphical Representation of Probabilities.

The Binomial distribution is a discrete probability distribution that models the number of successes in a fixed number of independent Bernoulli trials, each with the same probability of success.

Binomial Distribution Formula

$P(X = k) = C(n, k) \cdot p^k \cdot (1 - p)^{n-k}$

Where:

n = number of trials

k = number of successes

p = probability of success on a single trial

$C(n, k) = n! / (k! \cdot (n-k)!)$

The Mean, Variance, and Standard Deviation of the Binomial Distribution are:

Mean: $\mu = E(X) = np$

Variance: $\sigma^2 = V(X) = np(1-p)$

Standard Deviation: $\sigma = \sqrt{(np(1-p))}$

How to Implement in Excel

= BINOM.DIST(k, n, p, FALSE)

Source of data: Theoretical calculation using Binomial PMF for n = 8, p = 0.8 / 0.5 / 0.2

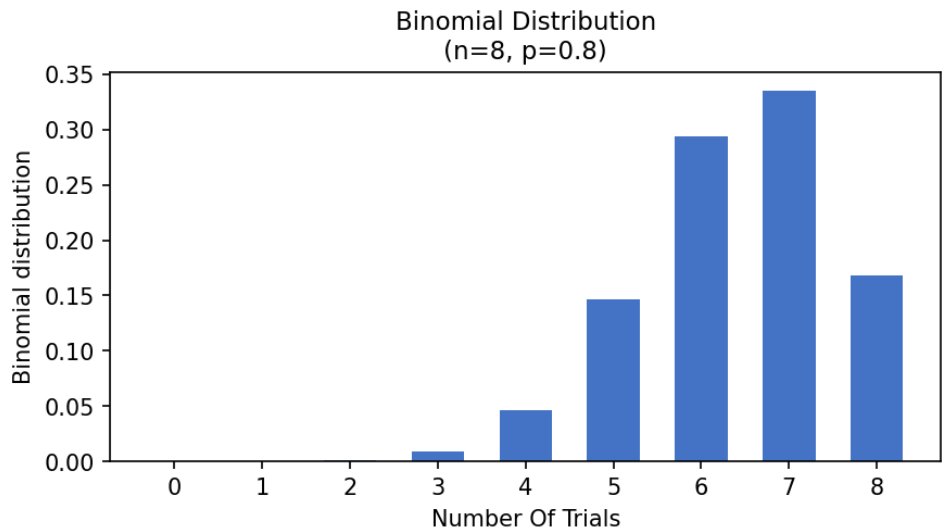
Note: When p is large (e.g., p = 0.8), the bulk of the probability falls in the larger numbers and the distribution tails off to the left.

Case 1: n = 8, p = 0.8

n (Number of Trials)	8
p (Probability of Success)	0.8

k (Number of Success)	**Binomial Distribution**
0	0.00000256
1	0.00008192
2	0.00114688
3	0.00917504

4	0.04587520
5	0.14680064
6	0.29360128
7	0.33554432
8	0.16777216

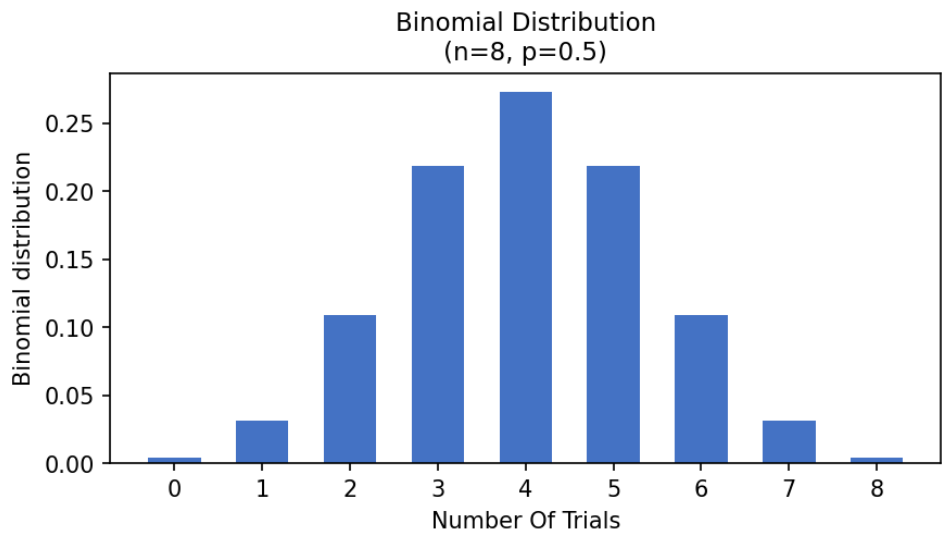


Case 2: n = 8, p = 0.5

n (Number of Trials)	**p (Probability of Success)**
8	0.5

k (Number of Success)	**Binomial Distribution**
0	0.00390625
1	0.03125000
2	0.10937500
3	0.21875000
4	0.27343750
5	0.21875000

6	0.10937500
7	0.03125000
8	0.00390625

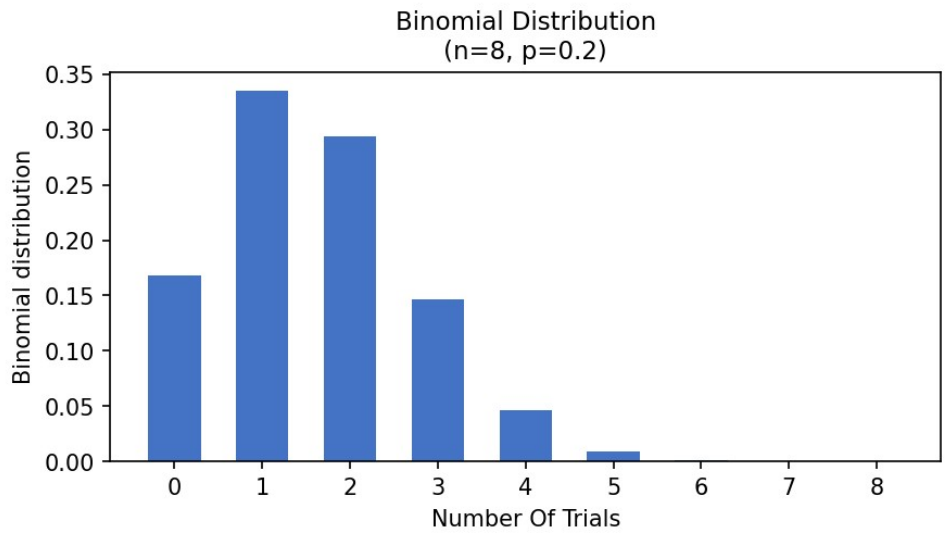


Case 3: n = 8, p = 0.2

n (Number of Trials)	**p (Probability of Success)**
8	0.2

k (Number of Success)	**Binomial Distribution**
0	0.16777216
1	0.33554432
2	0.29360128
3	0.14680064
4	0.04587520
5	0.00917504
6	0.00114688
7	0.00008192

8	0.00000256



Observation: When p = 0.5 the distribution is symmetric. When p>0.5 the distribution is left-skewed and when p<0.5 the distribution is right-skewed.

2. Plotting and Fitting of Multinomial Distribution and Graphical Representation of Probabilities.

The multinomial distribution is a multivariate generalization of the binomial distribution. Consider a trial that results in exactly one of some fixed finite number k of possible outcomes, with probabilities $p_1, p_2, \dots, p_k$ (so that $p_i \geq 0$ for $i = 1, \dots, k$ and $\Sigma_i p_i = 1$), and there are n independent trials. Then let the random variables X_i indicate the number of times outcome number i was observed over the n trials. Then $X = (X_1, X_2, \dots, X_k)$ follows a multinomial distribution with parameters n and p, where $p = (p_1, p_2, \dots, p_k)$.

Multinomial Distribution Formula

$p(x_1, x_2, \dots, x_k) = [n! / (x_1! \cdot x_2! \cdot \dots \cdot x_k!)] \cdot p_1\text{^}x_1 \cdot p_2\text{^}x_2 \cdot \dots \cdot p_k\text{^}x_k$

$Cov(X_i, X_j) = -np_ip_j \quad (i \neq j)$

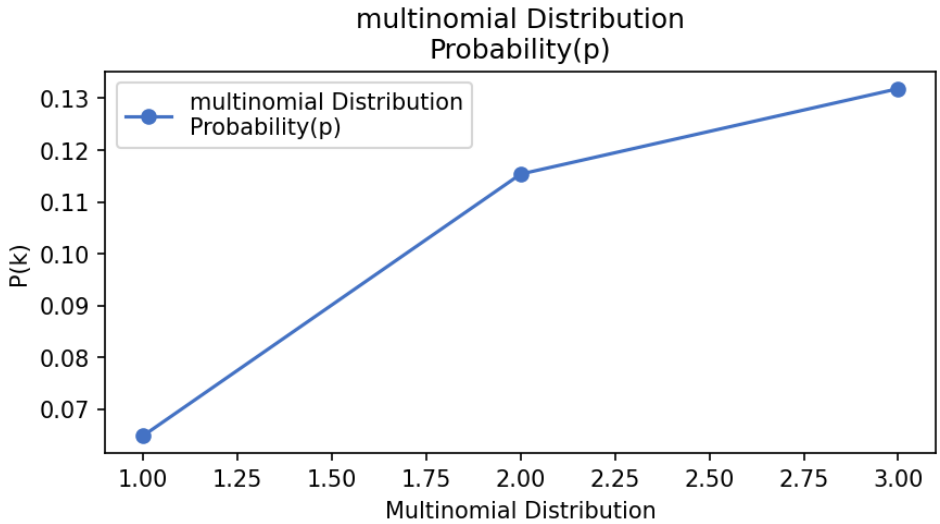
When $X = (x_1, x_2, \dots, x_k)$ follows a multinomial distribution with the PMF given above, X_i follows a binomial distribution with n trials and success probability p_i.

how to implement in excel

Multinomial = MULTINOMIAL(X1, X2, X3)

Probability = MULTINOMIAL * PRODUCT(p1^X1, p2^X2, p3^X3)

Parameter	1st Case	2nd Case	3rd Case
Red (X1)	4	3	2
Green (X2)	0	1	1
Blue (X3)	6	3	2
P_Red (P1)	0.375	0.375	0.375
P_Green (P2)	0.125	0.125	0.125
P_Blue (P3)	0.5	0.5	0.5
Multinomial	210	140	30
P1^X1	0.019775391	0.052734375	0.140625
P2^X2	1	0.125	0.125
P3^X3	0.015625	0.125	0.25
Probability	0.064888	0.115356445	0.131835938



Observation: The multinomial distribution probability decreases as the number of outcomes (multinomial coefficient) decreases across cases.

3. Plotting and Fitting of Poisson Distribution and Graphical Representation of Probabilities.

The Poisson distribution is a type of discrete probability distribution that determines the likelihood of an event occurring a specific number of times (k) within a designated time or space interval. This distribution is characterized by a single parameter, λ (lambda), representing the average number of occurrences of the event.

Poisson Distribution Formula

$P(k) = e^{(-\lambda)} \cdot \lambda^k / k!$

Statistic	Formula
Mean	$\mu = E(X) = \lambda$
Variance	$\sigma^2 = V(X) = \lambda$
Standard Deviation	$\sigma = \sqrt{\lambda}$

Where:

P(X = k) is the probability of observing k events

e is the base of the natural logarithm (approximately 2.71828)

λ is the mean number of successes that occur during a specific interval; λ = np

k is the number of successes

how to implement in excel

POISSON.DIST(number_s, average, cumulative)

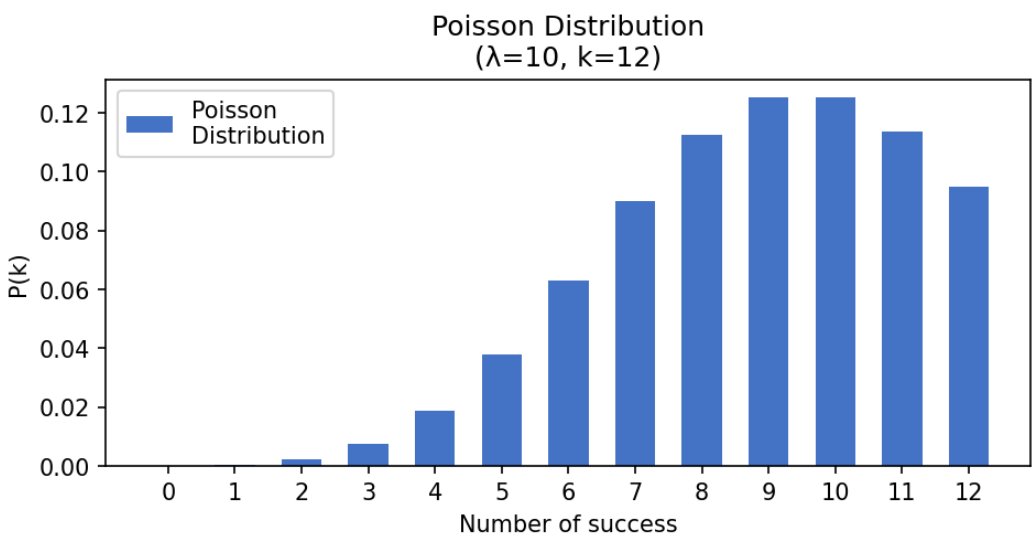
POISSON.DIST(k, λ, FALSE)

Data: λ = 10 (mean number of successes), k = 0 to 12

λ (Mean Number of Success)	10

k (Number of Success)	Poisson Distribution P(k)
0	4.53999E-05
1	0.000453999
2	0.002269996
3	0.007566655

4	0.018916637
5	0.037833275
6	0.063055458
7	0.090079226
8	0.112599033
9	0.125110036
10	0.125110036
11	0.113736396
12	0.094780330



Observation: The Poisson distribution peaks at k = λ. For λ = 10, the maximum probability is at k = 9 or k = 10.

4. Plotting and Fitting of Geometric Distribution and Graphical Representation of Probabilities.

In a Bernoulli trial, the likelihood of the number of successive failures before success is obtained is represented by a geometric distribution, which is a sort of discrete probability distribution. A Bernoulli trial is a test that can only have one of two outcomes: success or failure. In other words, a Bernoulli trial is repeated until success is obtained and then stopped in geometric distribution.

A geometric distribution is a discrete probability distribution that indicates the likelihood of achieving one's first success after a series of failures. The number of attempts in a geometric distribution can go on indefinitely until the first success is achieved. Geometric distributions are probability distributions that are based on three key assumptions.

- The trials that are being undertaken are self-contained.
- Each trial may only have one of two outcomes: success or failure.
- For each trial, the success probability, represented by p, is the same

Geometric Distribution formula

$$P(X = k) = (1 - p)^k \cdot p$$

Statistic	Formula
Mean:	$\mu = E(X) = 1/p$
Variance:	$\sigma^2 = V(X) = (1-p) / p^2$

where:

- k: number of failures before first success
- p: probability of success on each trial

The chance of a trial's success is denoted by p, whereas the likelihood of failure is denoted by q, q = 1 – p in this case. X ~ G(p) represents a discrete random variable, X, with a geometric probability distribution.

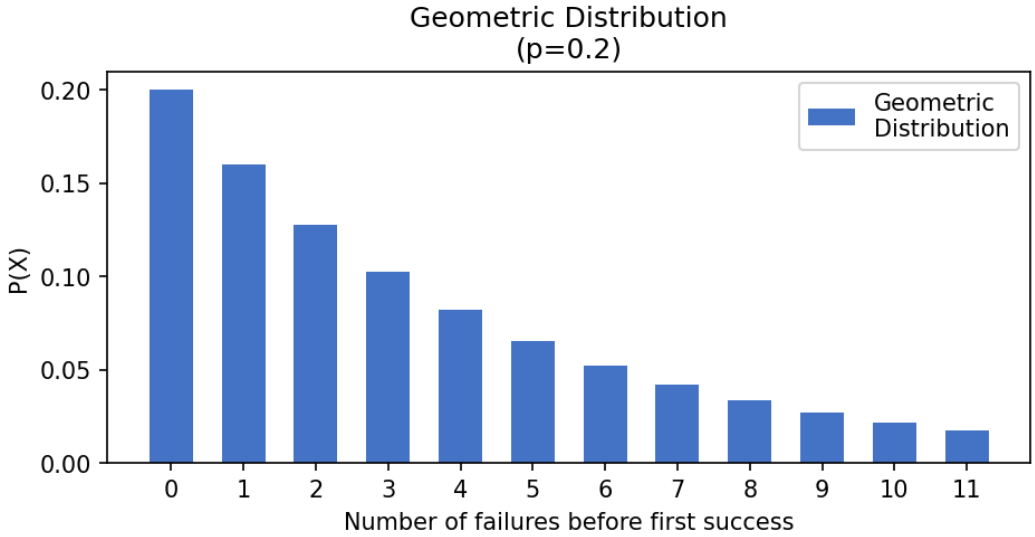
how to implement in excel

Probability = (1-p)^k * p

Prob. Of Success on Given trial(p)	0.2

Number of failures before first success	Geometric Distribution
0	0.2

1	0.16
2	0.128
3	0.1024
4	0.08192
5	0.065536
6	0.0524288
7	0.04194304
8	0.033554432
9	0.026843546
10	0.021474836
11	0.017179869



5. Plotting and Fitting of Uniform Distribution and Graphical Representation of Probabilities.

A uniform distribution is a distribution that has constant probability due to equally likely occurring events. It is also known as rectangular distribution (continuous uniform distribution). It has two parameters a and b: a = minimum and b = maximum. The distribution is written as U (a, b).

A uniform distribution is a type of probability distribution whereevery possible outcome has an equal probability of occurring. This means that all values within a given range are equally likely to be observed.

Uniform Distribution Formula

The probability density function (PDF) of a continuous uniform distribution defines the probability of a random variable falling within a particular interval. For a continuous uniform distribution over the interval [a,b].

$f(x) = 1 / (b - a) \quad \text{for } a \leq x \leq b$

Mean $\mu = (a + b) / 2$

Variance $\sigma^2 = (b - a)^2 / 12$

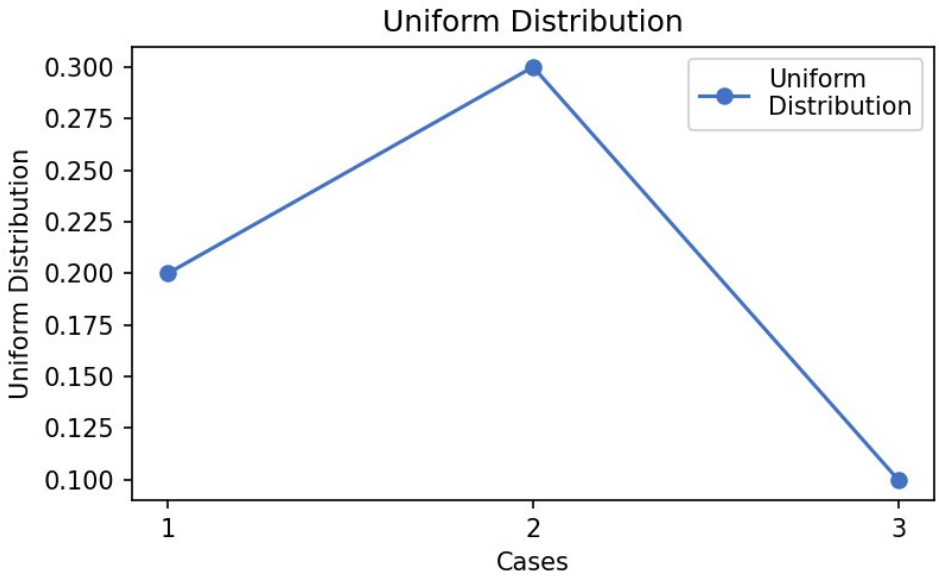
how to implement in excel

$\boldsymbol{P = (x_2 - x_1) / (b - a)}$

For calculating probability, we need:

1. *a: minimum value in the distribution*
2. *b: maximum value in the distribution*
3. x_1*: the minimum value you're interested in*
4. x_2*: the maximum value you're interested in*

Parameter	case 1	case 2	case3
a	15 (min)	15 (min)	15 (min)
b	25 (max)	25 (max)	25 (max)
x1	16	19	23
x2	18	22	24
Uniform Distribution	0.2	0.3	0.1


Uniform Distribution
Uniform Distribution
0.300
0.275
0.250
0.225
0.200
0.175
0.150
0.125
0.100
1
2
3
Cases
Uniform
Distribution

6. Plotting and Fitting of Exponential Distribution and Graphical Representation of Probabilities.

The support (set of values the Random Variable can take) of an Exponential Random Variable is the set of all positive real numbers. Suppose we are posed with the question- How much time do we need to wait before a given event occurs? The answer to this question can be given in probabilistic terms if we model the given problem using the Exponential Distribution. Since the time we need to wait is unknown, we can think of it as a Random Variable. If the probability of the event happening in a given interval is proportional to the length of the interval, then the Random Variable has an exponential distribution. The support (set of values the Random Variable can take) of an Exponential Random Variable is the set of all positive real numbers.

This distribution can be used to solve following type of real life problems-

- How long does a shop owner need to wait until a customer enter a shop.
- How long will a battery continue to work before it dies.
- How long will a computer continue to work before it breakdown.

$f(x) = \lambda e^{(-\lambda x)}, \quad x \geq 0$

$\quad 0, \qquad x < 0$

$E(X) = 1/\lambda, \quad Var(X) = 1/\lambda^2$

Here λ is the rate parameter and its effects on the density function.

e is a constant roughly equal to 2.718

How to Implement in excel

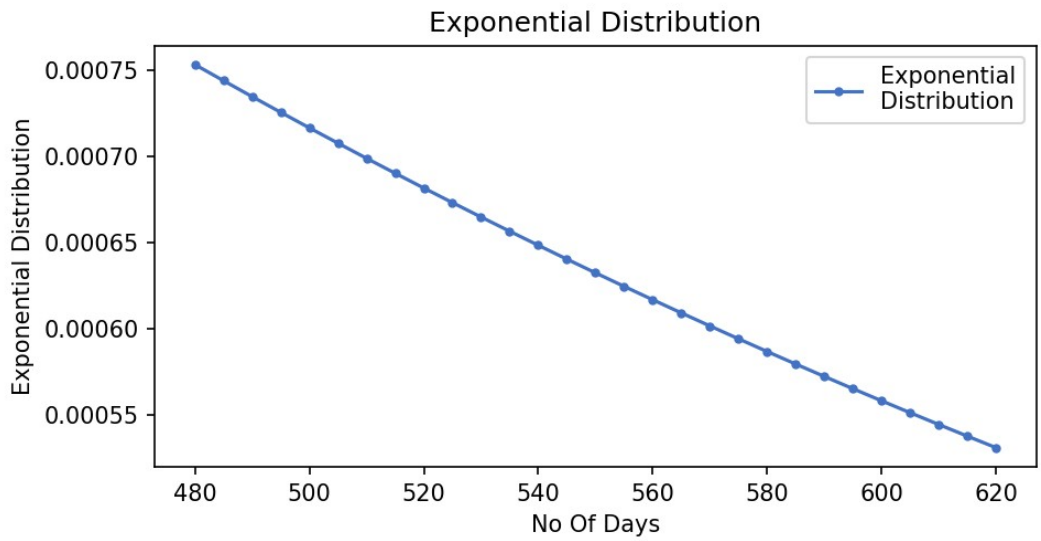
EXPON.DIST(X, lambda, cumulative)

EXPON.DIST(X, lambda, FALSE)

Parameter	Value
μ (Mean)	400
λ (Lambda)	0.0025

x	case 1	case 2	case 3
x	500	550	600
Exponential Distribution	0.000716	0.000632	0.000558

No Of Days	Exponential Distribution
500	0.000716
550	0.000632
600	0.000558



7. Plotting and Fitting of Normal Distribution and Graphical Representation of Probabilities.

We define Normal Distribution as the probability density function of any continuous random variable for any given system. Now for defining Normal Distribution suppose we take f(x) as the probability density function for any random variable X.

$f(x) \geq 0 \quad x \in (-\infty, +\infty),$

$f(x) = [1 / (\sigma\sqrt{(2\pi)})] \cdot e^{(-½ \cdot ((x-\mu)/\sigma)^2)}$

where,

- x is Random Variable
- μ is Mean
- σ is Standard Deviation

Properties of Normal Distribution

- For normal distribution of data, mean, median, and mode are equal, (i.e., Mean = Median = Mode).
- Total area under the normal distribution curve is equal to 1.
- Normally distributed curve is symmetric at the center along the mean.
- In a normally distributed curve, there is exactly half value to the right of the central and exactly half value to the right side of the central value.
- Normal distribution is defined using the values of the mean and standard deviation.
- Normal distribution curve is a Unimodal Curve, i.e. a curve with only one peak

how to implement in excel

1. Input your data set into an Excel spreadsheet

2. Find the mean of your data set

=AVERAGE(cell range)

- "cell range"is a required component and the range of cells where your data exists, such as cells A1 through A64. You can write this in the function as A1:A64.

3. Find the standard deviation of your data set

=STDEV(cell range)

4. Select a value for the distribution

5. Type the NORM.DIST function and fill

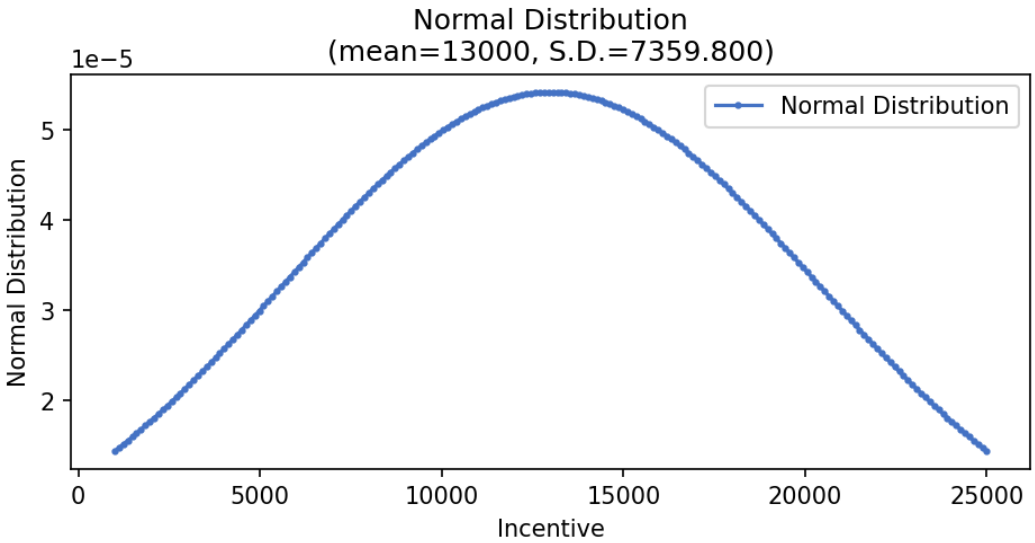
NORM.DIST(x, mean, standarddeviation, cumulative)

NORM.DIST(x, mean, standarddeviation, FALSE)

Employee	Incentive	Normal Distribution
EMP1	1000	1.43472E-05

EMP2	2000	1.77407E-05
EMP3	3000	2.15356E-05
EMP4	4000	2.56641E-05
EMP5	5000	3.00245E-05
EMP6	6000	3.44833E-05
EMP7	7000	3.88799E-05
EMP8	8000	4.3035E-05
EMP9	9000	4.6763E-05
EMP10	10000	4.98843E-05
EMP11	11000	5.22406E-05
EMP12	12000	5.37075E-05
EMP13	13000	5.42056E-05
EMP14	14000	5.37075E-05
EMP15	15000	5.22406E-05
EMP16	16000	4.98843E-05
EMP17	17000	4.6763E-05
EMP18	18000	4.3035E-05
EMP19	19000	3.88799E-05
EMP20	20000	3.44833E-05
EMP21	21000	3.00245E-05
EMP22	22000	2.56641E-05
EMP23	23000	2.15356E-05
EMP24	24000	1.77407E-05
EMP25	25000	1.43472E-05

	Value
mean	13000
standard Deviation	7359.800722



8. Calculation of Cumulative Distribution Functions for Exponential and Normal Distribution.

The cumulative distribution function (CDF) gives the probability that a random variable X takes a value less than or equal to x. For both Normal and Exponential distributions, Excel provides direct CDF computation.

For Normal CDF: NORM.DIST(x, mean, std_dev, TRUE)

For Exponential CDF: EXPON.DIST(x, lambda, TRUE)

Normal Distribution CDF

Employee	Incentive	Normal Distribution (CDF)
EMP1	1000	0.572137292
EMP2	2000	0.641935216
EMP3	3000	0.707279533
EMP4	4000	0.766470549
EMP5	5000	0.818348930
EMP6	6000	0.862343557
EMP7	7000	0.898442582
EMP8	8000	0.927102430
EMP9	9000	0.949118248
EMP10	10000	0.965881826

	Value
mean	5500
standard Deviation	3027.650354

Exponential Distribution CDF

No. of Days	Exponential CDF
300	0.527633447
320	0.550671036

350	0.583813798
400	0.632120559
420	0.650062251
450	0.675347533
500	0.713495203
520	0.727468207
550	0.747160404
600	0.776869840

	Value
λ	0.0025

Observation: The CDF is a monotonically increasing function bounded between 0 and 1. As $x \to \infty$, CDF $\to$ 1. Both Normal and Exponential CDFs demonstrate this property.

9. Given Data from Two Distributions, Find the Distance Between the Distributions.

<u>Euclidean Distance</u>

Euclidean distance is the distance between two real distinct value .It is calculate by the square root of the sum of the squared difference elements in two vectors.

Euclidean Distance = $|X - Y| = \sqrt{[\Sigma_{i=1}^{n} (x_i - y_i)^2]}$

X: Array or vector X

Y: Array or vector Y

x_i: Values of horizontal axis in the coordinate plane

y_i: Values of vertical axis in the coordinate plane

n: Number of observations

how to implement in excel

= SORT(SUM X MYZ(array_X, array_Y))

Parameters: n = 0.4, k = 8, λ = 0.4

Trials	Binomial Distribution	Poisson Distribution	distance
1	0.08957952	0.268128018	0.178548
2	0.20901888	0.053625604	0.155393
3	0.27869184	0.00715008	0.271542
4	0.23224320	0.000715008	0.231528
5	0.12386304	5.72006E-05	0.123806
6	0.04128768	3.81338E-06	0.041284
7	0.00786432	2.17907E-07	0.007864
8	0.00065536	1.08954E-08	0.000655

Observation: The Euclidean distance between Binomial (n=8, p=0.4) and Poisson (λ=0.4) distributions is largest at Trial 3, reflecting the divergence of these distributions in the middle range.

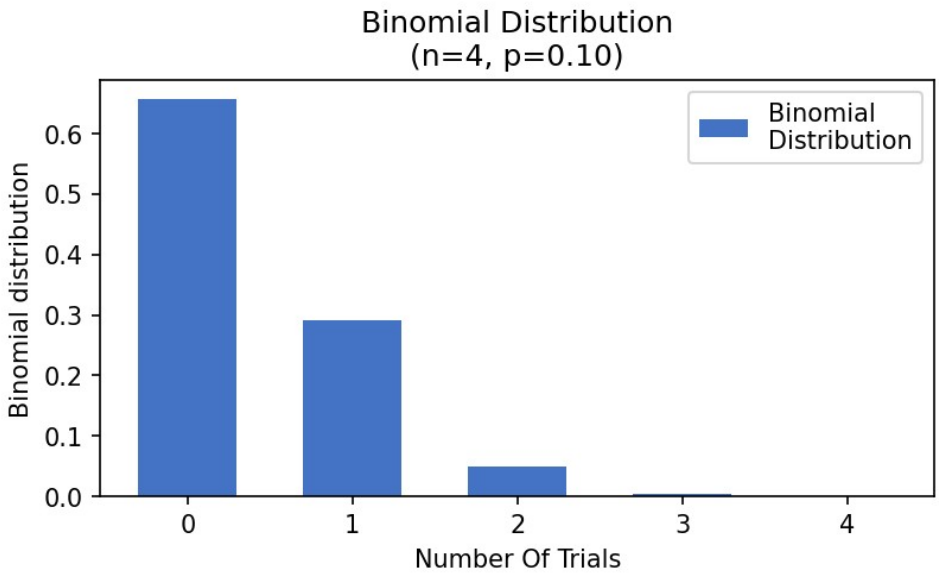
10. Application Problems Based on the Binomial Distribution.

Problem Statement

Antibiotics occasionally cause nausea as a side effect. A major drug company has developed a new antibiotic called Phe-Mycin. The company claims that, at most, 10 percent of all patients treated with Phe-Mycin would experience nausea as a side effect of taking the drug. Suppose that we randomly select n = 4 patients and treat them with Phe-Mycin. Each patient will either experience nausea (which we arbitrarily call a success) or will not experience nausea (a failure). We will assume that p, the true probability that a patient will experience nausea as a side effect, is .10, the maximum value of p claimed by the drug company. Furthermore, it is reasonable to assume that patients'reactions to the drug would be independent of each other. Let x denote the number of patients among the four who will experience nausea as a side effect. It follows that x is a binomial random variable, which can take on any of the potential values 0, 1, 2, 3, or 4. That is, anywhere between none of the patients and all four of the patients could potentially experience nausea as a side effectSuppose we wish to investigate whether p, the probability that a patient will experience nausea as a side effect of taking Phe-Mycin, is greater than.10, the maximum value of p claimed by the drug company. This assessment will be made by assuming, for the sake of argument, that p equals .10, and by using sample information to weigh the evidence against this assumption and in favor of the conclusion that p is greater than .10. Suppose that when a sample of n=4 randomly selected patients is treated with Phe-Mycin, three of the four patients experience nausea. Because the fraction of patients in the sample that experience nausea is 3/4 = .75, which is far greater than .10, we have some evidence contradicting the assumption that p equals .10.

n (Number of Trials)	p (Probability of Success on Given Trials)
4	0.1

k (Number of Success)	Binomial Distribution
0	0.6561
1	0.2916
2	0.0486
3	0.0036
4	0.0001



Formula: =BINOM.DIST(k, 4, 0.1, FALSE)

Conclusion: $P(X \geq 3) = P(X=3) + P(X=4) = 0.0036 + 0.0001 = 0.0037$. Since this probability is very small (0.37%), it provides strong evidence that $p>0.10$, contradicting the drug company's claim.

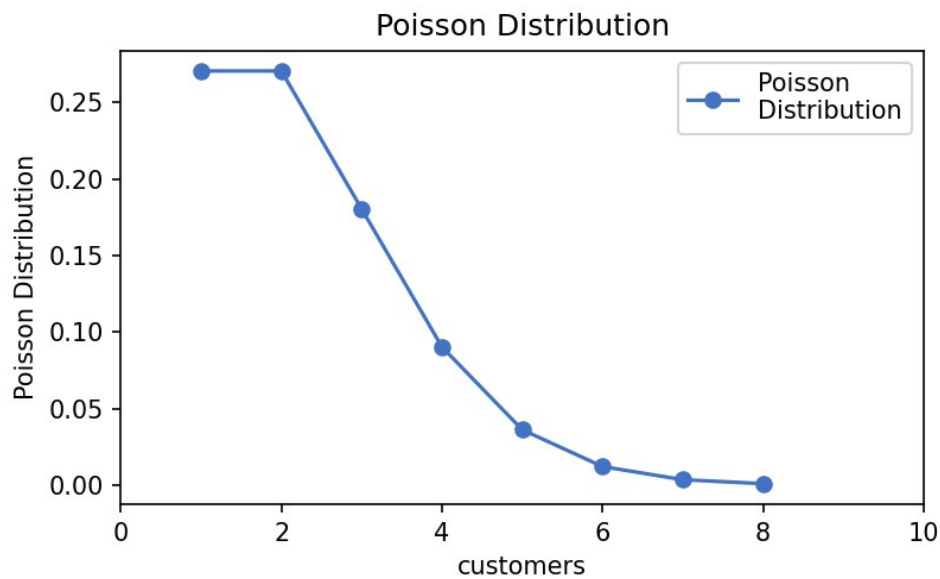
11. Application Problems Based on the Poisson Distribution.

Problem Statement

In a cafe, the customer arrives at a mean rate of 2 per min. Find the probability of arrival of 5 customers in 1 minute using the Poisson distribution formula.

Parameter	Value
λ (Mean Rate)	2
x (Target)	5

x (Number of Customers)	Poisson Distribution P(X = x)
1	0.270670566
2	0.270670566
3	0.180447044
4	0.090223522
5	0.036089409
6	0.012029803
7	0.003437087
8	0.000859272



Formula used: =POISSON.DIST(x, 2, FALSE)

Answer: The probability of exactly 5 customers arriving in 1 minute is $P(X = 5) \approx 0.0361$ or approximately 3.61%.

Observation: The distribution peaks at $x = 1$ and $x = 2$ (both ≈ 0.2707), which equals $\lambda = 2$, as expected for a Poisson distribution.

12. Application Problems Based on the Normal Distribution.

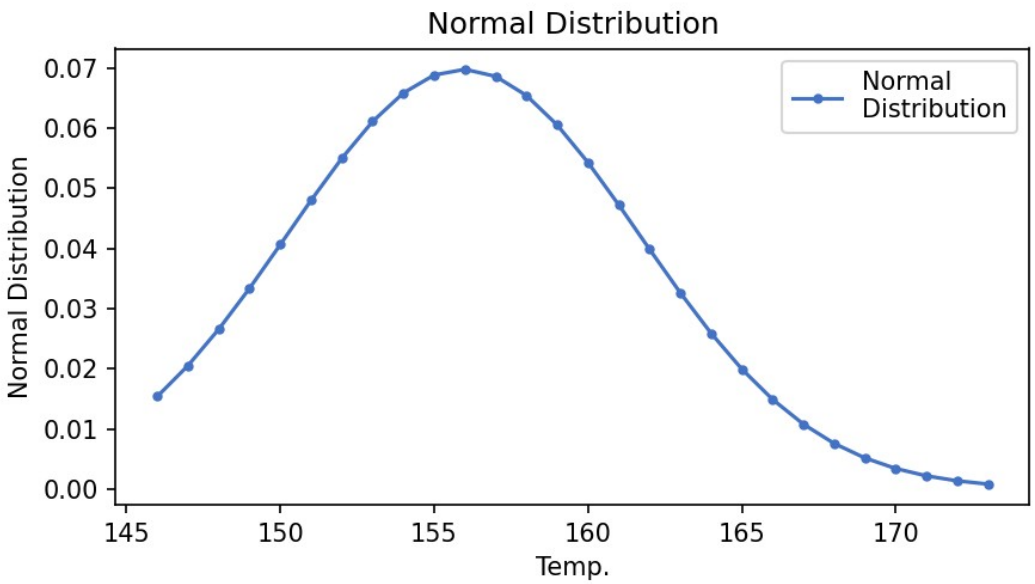
Problem Statement

According to the website of the American Association for Justice, ^11 Stella Liebeck of Albuquerque, New Mexico, was severely burned by McDonald's coffee in February 1992. Liebeck, who received third-degree bums over 6 percent of her body, was awarded $160,000 in compensatory damages and $480,000 in punitive damages. A post-verdict investigation revealed that the coffee temperature at the local Albuquerque McDonald's had dropped from about 185 degree F before the trial to about 158 degree after the trial. This case concerns coffee temperatures at a fast-food restaurant. Because of the possibility of future litigation and to possibly improve the coffee's taste, the restaurant wishes to study the temperature of the coffee it serves. To do this, the restaurant personnel measure the temperature of the coffee being dispensed (in degrees Fahrenheit) at a randomly selected time during each of the 24 half-hour periods from 8 a.m. to 7:30 p.m on a given day. This is then repeated on a second day, giving the 48 coffee temperatures in excel..

Temp.	Normal Distribution	Temp.	Normal Distribution
146	0.015374765	166	0.014888007
148	0.026566325	167	0.010783701
149	0.033356691	168	0.007575753
150	0.040621980	169	0.005161910
151	0.047980623	171	0.002186551
152	0.054966405	173	0.000819625
153	0.061073870	—	—
154	0.065817321	—	—
155	0.068794167	—	—
156	0.069741244	—	—
157	0.068573201	—	—
158	0.065395191	—	—
159	0.060487252	—	—
160	0.054263594	—	—

161	0.047214991	—	—
162	0.039845376	—	—
163	0.032613890	—	—
164	0.025891304	—	—
165	0.019935719	—	—

	Value
mean	155.947
standard Deviation	5.72008



Formula: =NORM.DIST(Temp, 155.947, 5.72008, FALSE)

Conclusion: The coffee temperature follows a near-normal distribution centered around 155.947°F. The probability peaks near the mean and tapers symmetrically on both sides.

13. Presentation of bivariate data through scatter-plot diagrams and calculations of covariance.

Covariance measures the joint variability of two random variables. If the greater values of one variable mainly correspond with the greater values of the other variable, and the same holds for the lesser values, i.e., the variables tend to show similar behavior, the covariance is positive.

The following are key differences between covariance and correlation:

The data set should be same size with the same number of data points.

The data set should not be empty nor should the standard Deviation of the value equal.

$Cov(X,Y) = \Sigma[(x_i - \bar{x})(y_i - \bar{y})] / n$

X and Y are the sample mean of the two set of values and n is the sample size.

5. Covariance is measure to indicate the extent to which two random variable in tandem.
6. Correlation is the measure used to represent how strongly two random variable are strongly related to each other.
7. Covariance is nothing but a measure of correlation.
8. Correlation referred to the scaled form of covariance.
9. Covariance can ary between $-\infty$ to $+\infty$ and correlation range between -1 to +1.
10. Covariance indicate the direction of the linear relationship between variables.
11. Correlation on the other hand measure both the strength and direction of the linear relationship between two variables.
12. Covariance is affected by change in scale.
13. Correlation is not affected by the change in scale.

Pearson Correlation Coefficient formula

$r = \Sigma[(x_i - \bar{x})(y_i - \bar{y})] / \sqrt{[\Sigma(x_i - \bar{x})^2 \cdot \Sigma(y_i - \bar{y})^2]}$

r = correlation coefficient

x_i = values of the x-variable in a sample

$\bar{x}$ = mean of the values of the x-variable

y_i = values of the y-variable in a sample

$\bar{y}$ = mean of the values of the y-variable

= PEARSON(arry:arry2)

Scatter plots

Scatter plots are the graphs that present the relationship between two variables in a data-set. It represents data points on a two-dimensional plane or on a Cartesian system. The

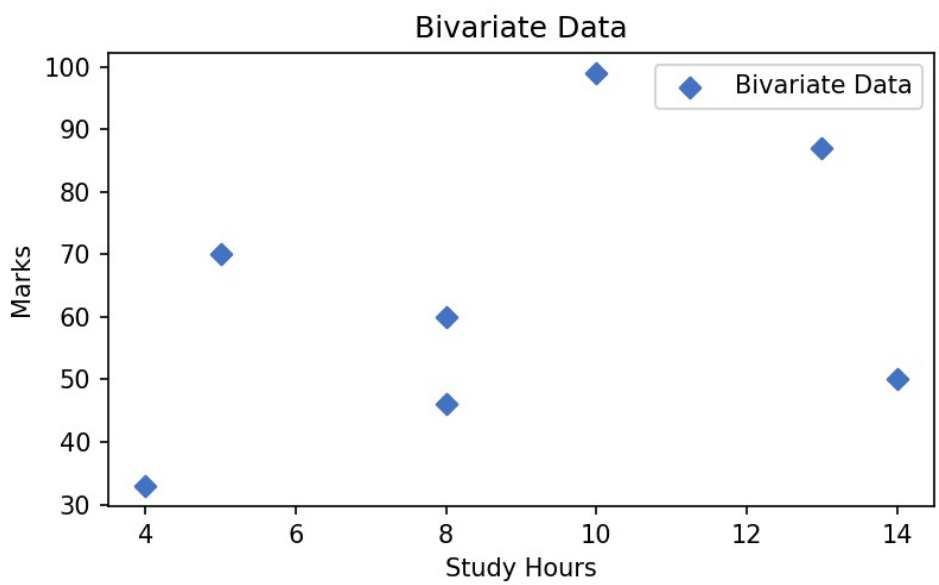
independent variable or attribute is plotted on the X-axis, while the dependent variable is plotted on the Y-axis. These plots are often called scatter graphs or scatter diagrams.

Scatter plots instantly report a large volume of data. It is beneficial in the following situations –

- For a large set of data points given
- Each set comprises a pair of values
- The given data is in numeric form

Study Hours	**10**	**13**	**14**	**8**	**5**	**4**	**8**

Marks	**99**	**87**	**50**	**60**	**70**	**33**	**46**



Statistic	**Value**
covariance	34.92857143

Click insert tab along the top ribbon then click scatter chart within chart group.

CORRELATION in excel and COVARIANCE in excel –

= CORREL(hours, score)

= COVARIANCE.P(hours, score)

14. Calculation of Karl Pearson's Correlation Coefficients.

x	Y1	Y2	Y3
2	80	65	10
5	95	69	30
6	76	60	15
8	58	95	25
10	67	80	10

Karl Pearson's correlation	Value
XY1	-0.62732
XY2	0.633225
XY3	0.018149

Interpretation:

XY1 = -0.627: A moderate negative correlation — as the study index increases, theory marks tend to decrease.

XY2 = +0.633: A moderate positive correlation — students with higher study indices score better in practicals.

XY3 = +0.018: An almost negligible correlation between study index and project marks.

15. To Find the Correlation Coefficient for a Bivariate Frequency Distribution.

Marks / Age In Years	16_18	18_20	20_22	22_24	Total
10_20	2	1	1	0	4
20_30	3	2	3	1	9
30_40	3	3	5	6	17
40_50	2	2	3	4	11
50_60	0	1	2	2	5
60_70	0	1	2	1	4

Marginal Frequency Distribution of X:	
Marks	Total
15	4
25	9
35	17
45	11
55	5
65	4

Age In Years	Total
17	10
19	10
21	16
23	14

Correlation Measure	Value
correlation coefficient along x-axis	-0.18773
correlation coefficient along Y-axis	0.774597

Observation: The correlation along the Y-axis (Age) is strongly positive (0.775), indicating that older students tend to score higher marks. The X-axis correlation is weakly negative.

16. Generating Random Numbers from Discrete (Bernoulli, Binomial, Poisson) Distributions.

How to implement in excel-

= BINOM.INV(1, P, RAND())

This function will generate 1 or 0 with chance of 1 being P random number

Random numbers from Binomial distributions	
n=10	
p=0.4	
	7
	4
	6
	4
	3

Note: These values are randomly generated and will differ each time the spreadsheet recalculates. The sample values shown above are representative of what would be generated with n = 10 and p = 0.4.

Bernoulli Distribution: A special case of Binomial with n = 1. Use =BINOM.INV(1, P, RAND()) which returns 1 (success) with probability P and 0 (failure) with probability 1-P.

17. Generating Random Numbers from Continuous (Uniform, Normal) Distributions.

= NORMINV(RAND(), B2, C2)

Where this RAND() function create your probability . B2 provides you mean , C2 refers your standand deviation.

Random numbers from Normal distributions	
mean=20	
standand Deviation=5	
	18.46540668
	19.13569112
	22.4762389
	21.37881651
	22.72814895

Uniform Distribution Random Numbers:

= a + (b - a) * RAND()

Where a = minimum value and b = maximum value of the uniform distribution. RAND() generates a uniform random number between 0 and 1, then scales it to [a, b].

Example: To generate uniform random numbers between 10 and 50:

= 10 + (50 - 10) * RAND()

Note: Since RAND() is volatile, random numbers recalculate every time the worksheet changes. To lock values, copy and paste as values only.

18. Find the Entropy from the Given Data Set.

The entropy of a random variable is the average level of information, surprise, or uncertainly inherent to the variable's possible outcomes. Given a discrete random random variable X which takes value in the alphabet x and distributed according to the P:x[0,1]. The entropy is H[X]

$$H(X) = -\Sigma\, p(x) \log_2 p(x) \qquad x \in X$$

The choice of base for log varies for different applications.

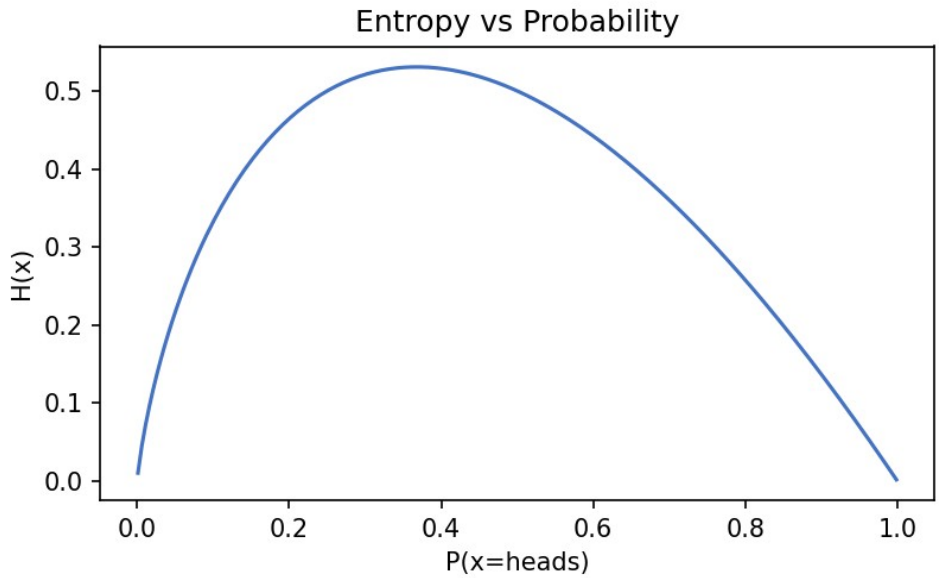
Base 2 gives the unit of bits while base e gives natural units.

Base o gives the units of H(X).

An equivalent definition of entropy is the expected value of the self information of a variable.

	P(x)	**log2p(x)**	**H(x)**
HH	1	0	0
HT	0.5	-1	0.5
TH	0.5	-1	0.5
TT	0	0	0

P(x)	**H(x)**
1	0
0.5	1
0	0



Total Entropy H(X) = 0 + 0.5 + 0.5 + 0 = 1.0 bit

Observation: Entropy is maximized when all outcomes are equally likely (P = 0.5 for binary gives H = 1 bit). Entropy is zero when the outcome is certain (P = 0 or P = 1).

Formula used in Excel: H(x) = -P(x) * LOG(P(x), 2) [for individual terms]

Total entropy: =SUMPRODUCT(-probabilities, LOG(probabilities, 2))